

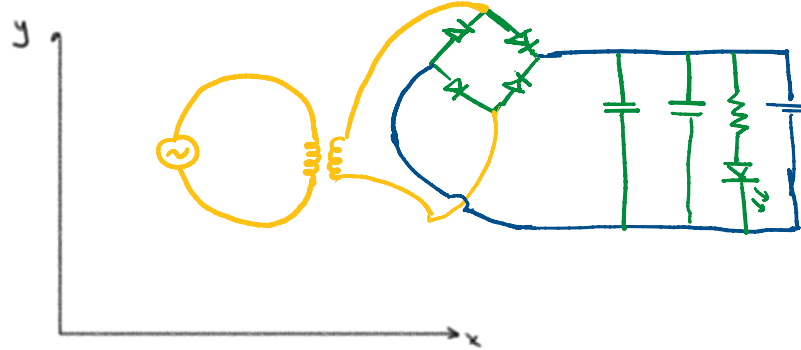
Arc Length

Thursday, May 8, 2025

9:39 AM

3. Length of a Curve.

3.1 How much wiring would this circuit need?:



If we know the parametrization of each curve, we can compute the arc length

3.2 Length of a Curve

(10)

$$l = \int_a^b \sqrt{\mathbf{r}' \cdot \mathbf{r}'} dt$$

$$\left(\mathbf{r}' = \frac{d\mathbf{r}}{dt} \right).$$

l is called the length of C , and C is called rectifiable.

3.3 Find the length of

Circular helix $\mathbf{r}(t) = [4 \cos t, 4 \sin t, 5t]$ from $(4, 0, 0)$ to $(4, 0, 10\pi)$.

Solution

Based on $z = 5t$ ranging from 0 to 10π , we deduce:

$$0 \leq t \leq 2\pi$$

From $\vec{r}(t) = [4 \cos t, 4 \sin t, 5t]$

get $\vec{r}'(t) = [-4 \sin t, 4 \cos t, 5]$

So that

$$\vec{r}'(t) \cdot \vec{r}'(t) = \begin{bmatrix} -4 \sin t \\ 4 \cos t \end{bmatrix} \cdot \begin{bmatrix} -4 \sin t \\ 4 \cos t \end{bmatrix} = (-4 \sin t)^2 + (4 \cos t)^2 + 5^2$$

So that

$$\begin{aligned}\vec{r}'(t) \cdot \vec{r}'(t) &= \begin{bmatrix} -4\sin t \\ 4\cos t \\ 5 \end{bmatrix} \cdot \begin{bmatrix} -4\sin t \\ 4\cos t \\ 5 \end{bmatrix} = (-4\sin t)^2 + (4\cos t)^2 + 5^2 \\ &= 16\sin^2 t + 16\cos^2 t + 25 \\ &= 16 + 25 = 41\end{aligned}$$

$$\text{Then } l = \int_0^{2\pi} \sqrt{\vec{r}' \cdot \vec{r}'} dt = \int_0^{2\pi} \sqrt{41} dt = 2\pi\sqrt{41}$$

1- Arc length Parametrization

4.1 Suppose C is parameterized by $\vec{r}(t)$ for $a \leq t \leq b$.
We create the arc-length function.

$$s(t) = \int_a^t \sqrt{\vec{r}'(u) \cdot \vec{r}'(u)} du$$

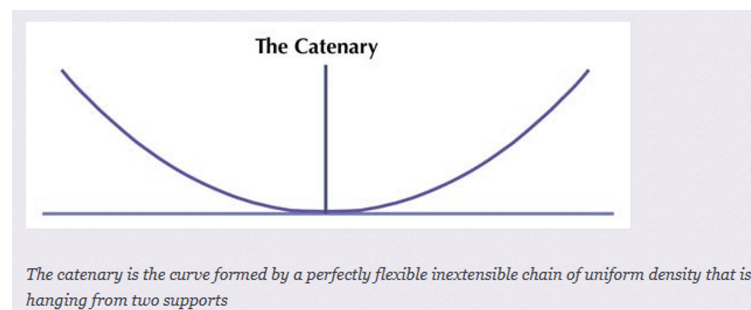
where u is a dummy variable.

$$s(t) = \int_a^t \sqrt{\mathbf{r}' \cdot \mathbf{r}'} d\tilde{t}$$

$$\left(\mathbf{r}' = \frac{d\mathbf{r}}{d\tilde{t}} \right).$$

4.2 Example

A Catenary



Curve C is given by: $\vec{r}(t) = [t, \cosh(t)]$ for $0 \leq t \leq 1$

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Note: $\frac{d}{dt} \cosh(t) = \sinh(t)$ [you are not responsible for hyperbolic functions]

Then $\vec{r}'(t) = \begin{bmatrix} 1 \\ \sinh(t) \end{bmatrix}$

so: $\vec{r}'(t) \cdot \vec{r}'(t) = 1 + \sinh^2(t) = \cosh^2(t)$

Thus, for $0 \leq t \leq 1$, the arclength function is:

$$\begin{aligned} s(t) &= \int_0^t \sqrt{\vec{r}'(w) \cdot \vec{r}'(w)} dw \\ &= \int_0^t \sqrt{\cosh^2(w)} dw \\ &= \int_0^t \cosh(w) dw = [\sinh(w)]_0^t \\ &= \sinh(t) - \sinh(0) \end{aligned}$$

$\therefore s(t) = \sinh(t)$

4.3 Arc-length Parameterization

The catenary $\vec{r}(t) = [t, \cosh(t)]$ can now be parameterized in terms of the arclength function $s(t) = \sinh(t)$ instead.

(i) $s = \sinh(t) \Rightarrow t = \sinh^{-1}(s)$ [a.k.a $\operatorname{arsinh}(s)$]

(ii) $1 + \sinh^2(t) = \cosh^2(t) \Rightarrow 1 + s^2 = \cosh^2(t) \Rightarrow \cosh(t) = \sqrt{s^2 + 1}$

(iii) as $0 \leq t \leq 1$, we find that $0 \leq s \leq \sinh(1)$

So the catenary can also be parameterized as

$$\vec{r}(s) = \begin{bmatrix} \sinh^{-1}(s) \\ \sqrt{s^2 + 1} \end{bmatrix}, \quad 0 \leq s \leq \sinh(1)$$

Looks Complicated! Why would any one do this?!?

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4.4 Of little practical value, but arclength parametrization has a lot of theoretical value.

For instance

$$\left. \begin{aligned} \frac{d}{ds} \sinh^{-1}(s) &= \frac{1}{\sqrt{s^2+1}} \\ \frac{d}{ds} (\sqrt{s^2-1}) &= \frac{s}{\sqrt{s^2+1}} \end{aligned} \right\} \vec{\tilde{r}}'(s) = \begin{bmatrix} \frac{1}{\sqrt{s^2+1}} \\ \frac{s}{\sqrt{s^2+1}} \end{bmatrix}$$

Then speed can be obtained from:

$$\begin{aligned} \vec{\tilde{r}}'(s) \cdot \vec{\tilde{r}}'(s) &= \left(\frac{1}{\sqrt{s^2+1}} \right)^2 + \left(\frac{s}{\sqrt{s^2+1}} \right)^2 \\ &= \frac{1}{s^2+1} + \frac{s^2}{s^2+1} \\ &= \frac{1+s^2}{s^2+1} = 1 \end{aligned}$$

Thus the speed is 1, and the velocity $\vec{\tilde{r}}'(s)$ is a unit vector!

4.4 Any curve C with arc-length parameterization $\vec{r}(s)$ always has unit-speed and thus the velocity $\vec{r}'(s)$ is a unit tangent vector.

Arc Length as Parameter. The use of s in (1) instead of an arbitrary t simplifies various formulas. For the unit tangent vector (8) we simply obtain

(14) $\mathbf{u}(s) = \mathbf{r}'(s).$

Indeed, $|\mathbf{r}'(s)| = (ds/ds) = 1$ in (12) shows that $\mathbf{r}'(s)$ is a unit vector. Even greater simplifications due to the use of s will occur in curvature and torsion (below).